UCLA Anderson
Course Project Presentation

Dynamic S&P 500 GNN Portfolio

Direct GNNs, RF + GCN features, and monthly OOS portfolio results

Project Group 6
WRDS/CRSP dynamic constituents | Monthly rebalancing | 2015–2024 OOS

Executive Answer

The final result is not "GNNs beat everything." It is more specific and more useful.

Stage 1

Pure GNN portfolios are unstable.

- GCN and GraphSAGE are competitive but do not clearly dominate EqualWeight.
- GAT and APPNP are more architecture-sensitive.

Stage 2

GCN features help RF.

- GCN score and embeddings add relational context.
- RF + GCN improves RF Base on return, Sharpe, and drawdown.

Stage 3

Final strategy is frozen.

- Six original features + GCN outputs.
- Five-seed RF ensemble.
- Long-only equal-weight top 20.

Final ann. return	Final Sharpe	Final max drawdown
12.33%	**0.676**	**-24.99%**

Dynamic constituents reduce survivor-only bias.

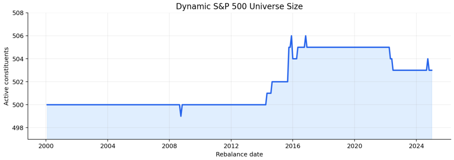
- WRDS/CRSP daily stock data and historical S&P 500 membership.
- Stocks are identified by PERMNO rather than ticker.
- At each rebalance, only active index constituents are investable.
- OOS: 2015-01-30 to 2024-11-29, 119 monthly rebalances.

Graph at month t

$$A_{ij,t} = \begin{cases} \rho_{ij,t}^{(60)}, & j \in \mathrm{TopK}_{10}^{+}(i,t), \\ 0, & \text{otherwise.} \end{cases}$$

Monthly universe logic

The investable set is rebuilt at each rebalance date, while the graph edges are rebuilt from the latest 60 daily returns.



Universe count from local WRDS/CRSP dynamic membership file.

Prediction Setup: Cross-Sectional Ranking

Node features

$$x_{i,t} = [\mathrm{Mom}_{20}, \mathrm{Mom}_{60}, \mathrm{Vol}_{20}, \mathrm{Vol}_{60}, \mathrm{VolChg}_{20}, \beta_{60}]$$

- **Momentum** $(20, 60)$: short- and medium-horizon trend.
- **Volatility** $(20, 60)$: recent and slower risk state.
- **Volume change** (20): trading attention and liquidity pressure.
- **Beta** (60): market exposure relative to the index proxy.
- All signals are rolling, standardized cross-sectionally, and known by date t.
- **Design:** compact 6F inputs limit overfitting; the GNN learns cross-stock dependence.

Forward-return target

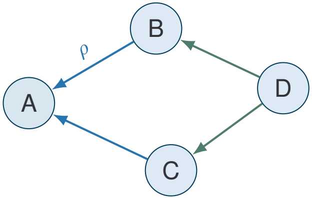
$$y_{i,t} = \frac{P_{i,t+21}}{P_{i,t+1}} - 1$$

- One-day execution lag avoids signal-close look-ahead.
- Monthly portfolio uses stock ranking, not calibrated return levels.
- Validation uses rank IC; portfolio rule uses top 20 selection.

Why ranking?

The trading rule only asks: which stocks are most attractive this month?

Common GNN View: Message Passing on a Stock Graph



$$h_i^{(\ell+1)} = \phi\left(W_{\text{self}}\, h_i^{(\ell)} + \sum_{j\in\mathcal{N}(i)} \omega_{ij,t}\, W_{\text{nbr}}\, h_j^{(\ell)}\right)$$

- $h_i^{(\ell)}$: representation of stock i.
- $\mathcal{N}(i)$: top correlated neighbors.
- $\omega_{ij,t}$: graph weight, learned attention, or normalized adjacency.

Core logic

- Build stock neighborhoods from positive rolling correlations.
- Aggregate neighbor features into stock i's representation.
- Use the updated representation for ranking and portfolio selection.

Financial interpretation

GNNs use cross-asset dependencies: sector, style, macro, and crowding effects.

Four GNN Models: Message-Passing Logic

GCN: normalized neighbor smoothing

$$H^{(\ell+1)} = \phi(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(\ell)}W^{(\ell)})$$

Stable, low-variance relational smoothing.

GraphSAGE: self + sampled neighbors

$$h_i^{(\ell+1)} = \phi(W_s h_i^{(\ell)} + W_n \mathrm{AGG}\{h_j^{(\ell)}\})$$

Flexible for changing node sets.

GAT: learned attention weights

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{r\in\mathcal{N}(i)}\exp(e_{ir})}, \qquad h_i' = \sum_j \alpha_{ij} W h_j$$

More expressive, but higher estimation risk.

APPNP: personalized PageRank propagation

$$H^{(r+1)} = (1-\alpha)\hat{A}H^{(r)} + \alpha Z$$

Diffuses MLP signal through the graph.

Model families: Kipf and Welling (2017), Hamilton et al. (2017), Velickovic et al. (2018), Klicpera et al. (2019).

Stage 1 Design: Direct GNN Portfolio Managers

Direct GNN Portfolio Pipeline

Node features X_t

↓

Dynamic correlation graph G_t

↓

GNN score $\hat{s}_{i,t}$

↓

Rank stocks

↓

Long top 20, equal weight

Test question Can a GNN score directly rank stocks well enough to construct a monthly long-only portfolio? **Training objective**

$$\mathcal{L} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)} \log(1 + \exp[-s_{ij}(\hat{s}_i - \hat{s}_j)])$$

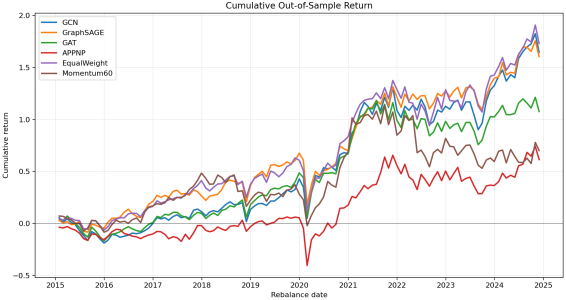
- Pairwise ranking loss.
- Validation: rank IC.
- OOS portfolios: monthly top 20.

Stage 1 Result: Pure GNNs Do Not Reliably Beat EqualWeight

Strategy	Return	Sharpe	Max DD	Turn.
GCN	10.31%	0.543	-26.54%	1.64
GraphSAGE	10.14%	0.591	-29.56%	1.66
GAT	7.65%	0.411	-28.62%	1.60
APPNP	4.95%	0.204	-43.88%	1.61
EqualWeight	10.66%	0.590	-31.48%	0.02
Momentum60	5.50%	0.273	-34.10%	1.29

Reading the table

GraphSAGE has competitive Sharpe, but high turnover; EqualWeight remains hard to dominate after costs.

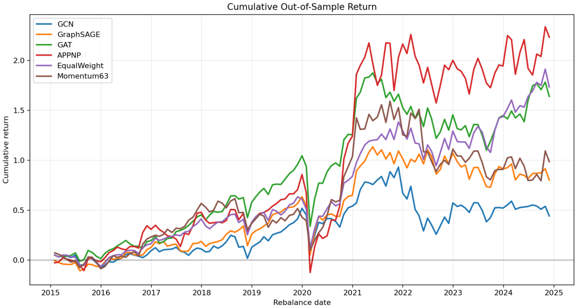


Robustness: More Features Did Not Fix Direct GNN Allocation

10F strategy	Return	Sharpe	Max DD
GCN	3.76%	0.193	-34.80%
GraphSAGE	6.11%	0.316	-35.60%
GAT	10.28%	0.531	-34.59%
APPNP	12.56%	0.430	-52.85%
EqualWeight	10.68%	0.592	-31.44%

Interpretation

- Richer inputs improved some architectures, but not consistently.
- Drawdowns remained unattractive.
- Direct GNN scores are sensitive to architecture and sample noise.



Why Direct GNN Allocation Is Difficult

Low signal-to-noise

Monthly stock returns are noisy, and each graph snapshot gives only one cross-section of labels.

Portfolio mismatch

A prediction loss can create rankings that turn over aggressively, while EqualWeight has almost no turnover.

Unstable relations

Correlation networks change through regimes; expressive attention may overfit short-lived relationships.

Stage 1 conclusion

Pure GNNs learn useful structure, but they are not reliable enough to serve as the final standalone portfolio engine in this sample.

This motivates changing the role of GNNs: use them to create relational representations, then let a tabular model perform the final ranking.

GCN-to-RF Feature Pipeline

Original six stock features

↓

Dynamic correlation graph

↓

GCN score +
64D embedding

↓

Random Forest rank model

↓

Top 20 portfolio

$$z_{i,t} = [x_{i,t} \| \hat{s}_{i,t}^{GCN} \| h_{i,t}^{GCN}]$$

$$\hat{s}_{i,t}^{RF} = f_{RF}(z_{i,t})$$

- GCN extracts relational context.
- RF handles nonlinear tabular interactions.
- This follows the spirit of relational stock ranking: graph information complements price features.

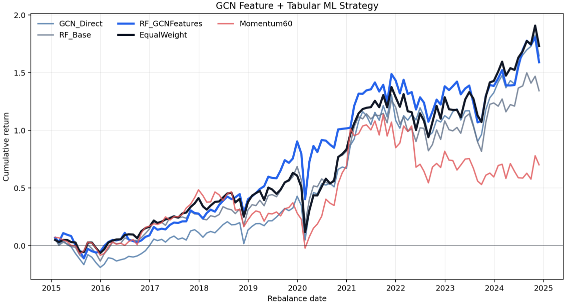
Related literature: Feng et al. (2019); Random Forest: Breiman (2001).

Stage 2 Evidence: GCN Features Improve RF

Strategy	Return	Sharpe	Max DD	Turn.
RF Base	8.97%	0.475	-29.79%	1.69
RF + GCN	10.07%	0.557	-26.32%	1.61
GCN Direct	10.31%	0.543	-26.54%	1.64
EqualWeight	10.66%	0.590	-31.48%	0.02
Momentum60	5.50%	0.273	-34.10%	1.29

Key comparison

RF + GCN raises Sharpe from 0.475 to 0.557 and reduces max drawdown by 3.47 percentage points versus RF Base.



Final Strategy: Frozen RF + GCN Pipeline

Six original features

↓

GCN score + hid-
den embeddings

↓

Five-seed RF ensemble

↓

Rank all active S&P
500 constituents

↓

Long top 20, equal-weight

↓

Subtract 10 bps
per unit turnover

What is frozen?

- Same six stock features.
- Same GCN feature extraction role.
- Same RF ensemble design.
- Same long-only top-20 rule.
- No annual model switching or after-the-fact rule selection.

$$\hat{s}_{i,t}^{ens} = \frac{1}{5}\sum_{m=1}^{5} \hat{s}_{i,t}^{(m)}$$

Final OOS Performance

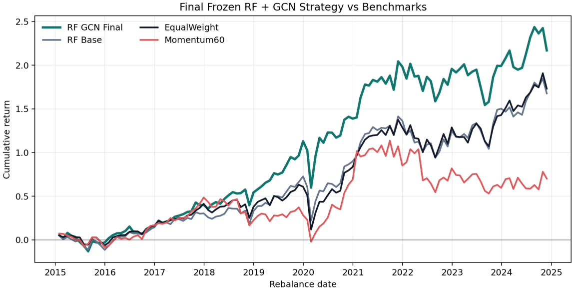
Strategy	Return	Sharpe	Max DD	Turn.
RF + GCN	12.33%	0.676	-24.99%	1.60
RF Base	10.44%	0.571	-29.25%	1.71
EqualWeight	10.66%	0.590	-31.48%	0.02
Momentum60	5.50%	0.273	-34.10%	1.29

Return lift
+1.89 pp

Sharpe lift
+0.105

MDD improvement
+4.26 pp

All improvements are measured versus RF Base.



Implementation Robustness Check

The alpha model is fixed; only the portfolio implementation rule changes.

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD	Turnover
RF + GCN Final	**12.33%**	**18.24%**	**0.676**	**-24.99%**	**1.60**
RF + GCN EqualTop20	12.35%	18.47%	0.668	-26.82%	1.61
RF + GCN BufferTop20	11.97%	18.23%	0.657	-25.78%	1.52
RF + GCN VolTop20	12.02%	18.07%	0.665	-26.59%	1.64
RF Base	10.44%	18.28%	0.571	-29.25%	1.71
EqualWeight	10.66%	18.06%	0.590	-31.48%	0.02
Momentum60	5.50%	20.14%	0.273	-34.10%	1.29

Same RF + GCN signal, monthly rebalancing, 2015–2024 OOS, and 10 bps transaction cost.

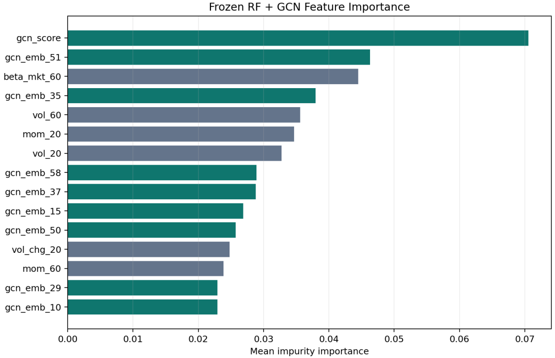
Variant suffixes

EqualTop20: select top 20 scores and equal-weight them.
BufferTop20: keep prior holdings while still inside the top 30 buffer.
VolTop20: top 20 names with inverse-volatility weights.

Takeaway

The frozen final rule gives the best Sharpe and shallowest drawdown among variants.

What the Final Model Learns



Top-3 importance

1. gcn_score **2.** gcn_emb_51 **3.** beta_mkt_60

GCN-derived variables survive inside the RF.

- The RF does not rely only on raw momentum and volatility.
- Several GCN embedding dimensions rank alongside traditional features.
- Interpretation: the graph representation adds information about a stock's network neighborhood.

Economic intuition

Similar-neighborhood stocks expose common risk and trend states; the RF uses this relational context as an additional signal.

Cost Sensitivity and Turnover

Cost	Strategy	Return	Sharpe	Max DD
0 bps	RF+GCN	14.49%	0.794	-24.66%
0 bps	EqualWeight	10.69%	0.592	-31.48%
10 bps	RF+GCN	12.33%	0.676	-24.99%
10 bps	EqualWeight	10.66%	0.590	-31.48%
20 bps	RF+GCN	10.22%	0.560	-25.32%
20 bps	EqualWeight	10.64%	0.589	-31.48%

Important limitation

- The strategy is active: average monthly turnover is about 1.60.
- At 10 bps, RF + GCN remains best on Sharpe and drawdown.
- At 20 bps, the active strategy's edge becomes fragile versus EqualWeight.

Implication

The model creates alpha-like rankings, but implementation costs matter.

Why GCN Is the Right Feature Extractor Here

Stable smoothing

GCN performs conservative neighborhood smoothing, which is valuable in low signal-to-noise return prediction.

RF compatibility

RF benefits from denoised graph embeddings, then handles nonlinear thresholds and feature interactions.

Lower overfit risk

Relative to attention-heavy GAT, GCN estimates fewer moving parts from noisy monthly labels.

Preferred interpretation

GCN is not selected because it is the most expressive architecture. It is selected because it offers the best balance of relational learning, stability, and interpretability for RF-based stock ranking.

Robustness Checks and Guardrails

What we checked

- Dynamic S&P 500 membership rather than current-member backfill.
- One-day execution lag in labels.
- 6F and 10F feature specifications.
- Direct GNNs versus RF + GCN hybrid.
- Final-stage refined RF features.
- Transaction cost sensitivity at 0, 10, and 20 bps.

What remains limited

- OOS is one historical path: 119 monthly rebalances.
- No shorting costs, borrow constraints, or market impact.
- Graph uses rolling correlations only.
- EqualWeight remains a difficult low-turnover benchmark.

Course-project positioning

The result is academic evidence for graph-aware representation value, not a production trading system.

Final Takeaways

1. **Direct GNN portfolios are not consistently superior.**
   They can rank stocks, but the portfolio results are architecture-sensitive and turnover-heavy.
2. **GCN outputs are useful features.**
   They inject relational stock-network information into RF, improving RF Base.
3. **The final strategy is frozen and interpretable.**
   Six features + GCN representation + RF seed ensemble + long-only top 20.
4. **Transaction costs are the main practical constraint.**
   At 10 bps the strategy works; at 20 bps the edge versus EqualWeight is fragile.

OOS months
119

Final ann. return
12.33%

Final Sharpe
0.676

Final max DD
-24.99%

Selected References

Graph neural networks

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Thank you